5.2.17

EE25BTECH11014 - Bhoomika Lokesh

Question: Solve the given system of linear equations:

$$x - 3y - 7 = 0 \tag{1}$$
$$3x - 3y - 15 = 0 \tag{2}$$

Solution:
The given equations of line can be expressed as:

$$\mathbf{n}^{\mathbf{T}}\mathbf{x} = c$$

$$\begin{pmatrix} 1 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 7 \tag{3}$$
$$\begin{pmatrix} 3 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 15 \tag{4}$$

The matrix we get by the above equations is as follows:

$$\begin{pmatrix} 1 & -3 \\ 3 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 15 \end{pmatrix} \tag{5}$$

The following is the augmented matrix which can be solved using Gaussian elimination.

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 3 & -3 & 15 \end{array}\right) \xrightarrow{R_2 \longrightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array}\right) \tag{6}$$
$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array}\right) \xrightarrow{R_2 \longrightarrow \frac{R_2}{6}} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array}\right) \tag{7}$$
$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow{R_1 \longrightarrow R_1 + 3R_2} \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -1 \end{array}\right) \tag{8}$$

From the matrix,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \tag{9}$$
$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \tag{10}$$

Hence the system of equations has a unique solution,$x = 4$ and $y = -1$.

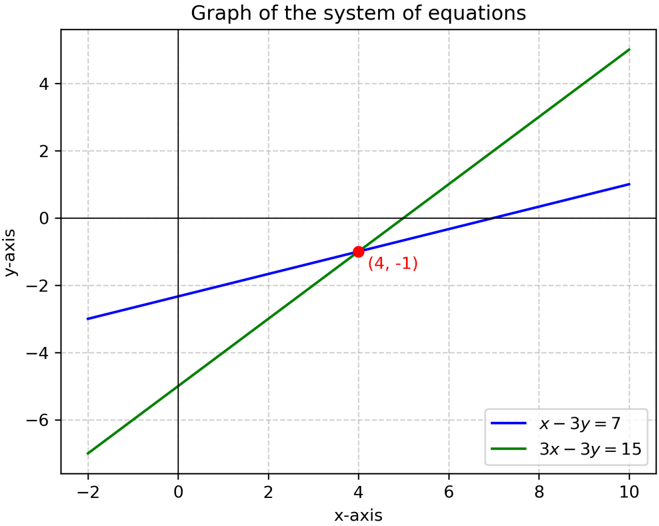

Fig. 0.1: System of Linear Equations